# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
## Department of Computer Science and Engineering
## ASSESSMENT TOOL 1 – EXPERIMENTS

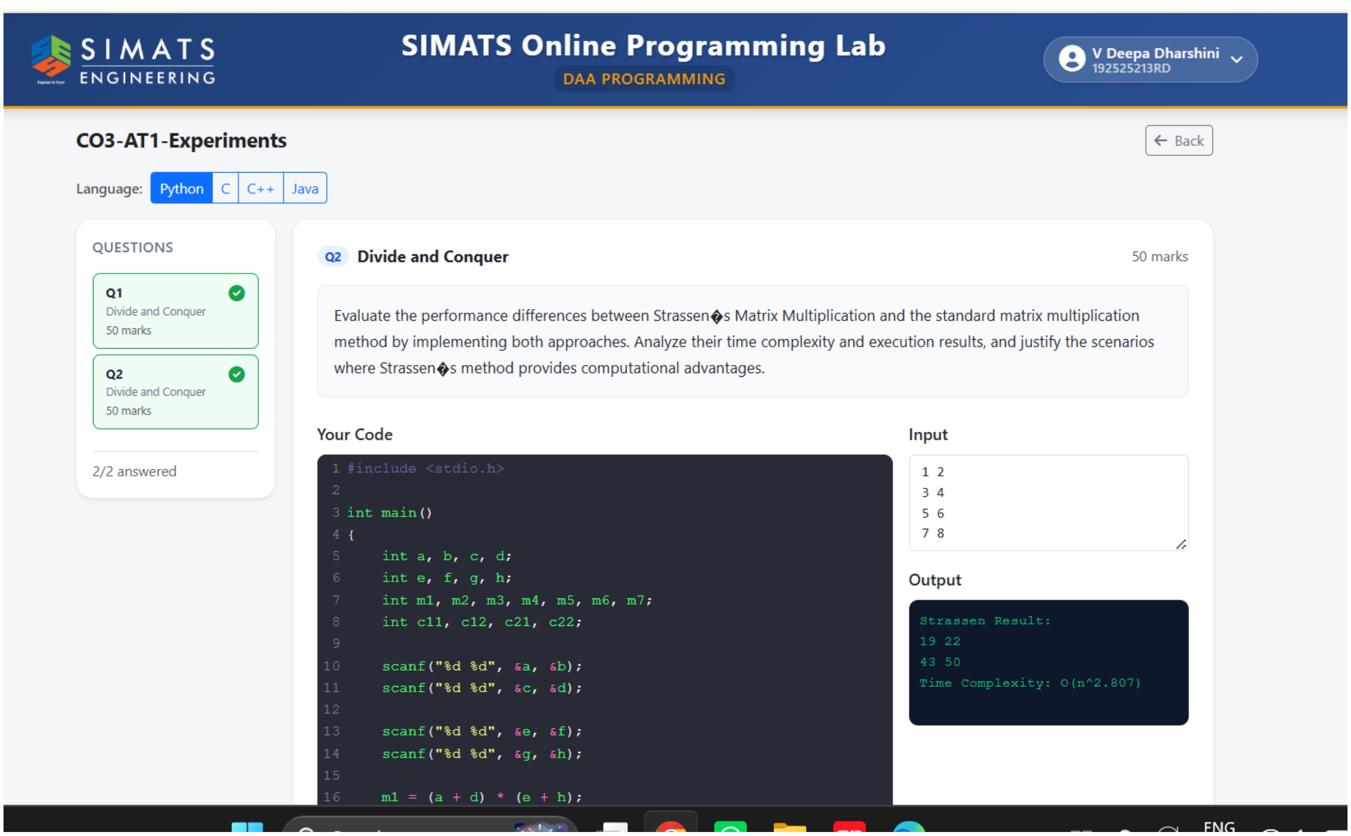
| | | | |
|---|---|---|---|
| **Course Code:** | CSA0621 | **Course Title:** | Design and Analysis of Algorithms |
| CO Assessed: | CO3: Experiments (BL4, BL5) | **Assessment:** | Assessment Tool 1 – Experiments |
| **Weightage:** | 40% of CIA | **Total Marks:** | 100 (2 Questions × 50 Marks) |
| **CO3 Statement:** | Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication | | |
| **Student Name:** | V Deepa Dharshini | **Reg. No.:** | 192525213 |
| **Section / Batch:** | B tech AIML(2nd year) | **Date:** | 13-08-2026 |

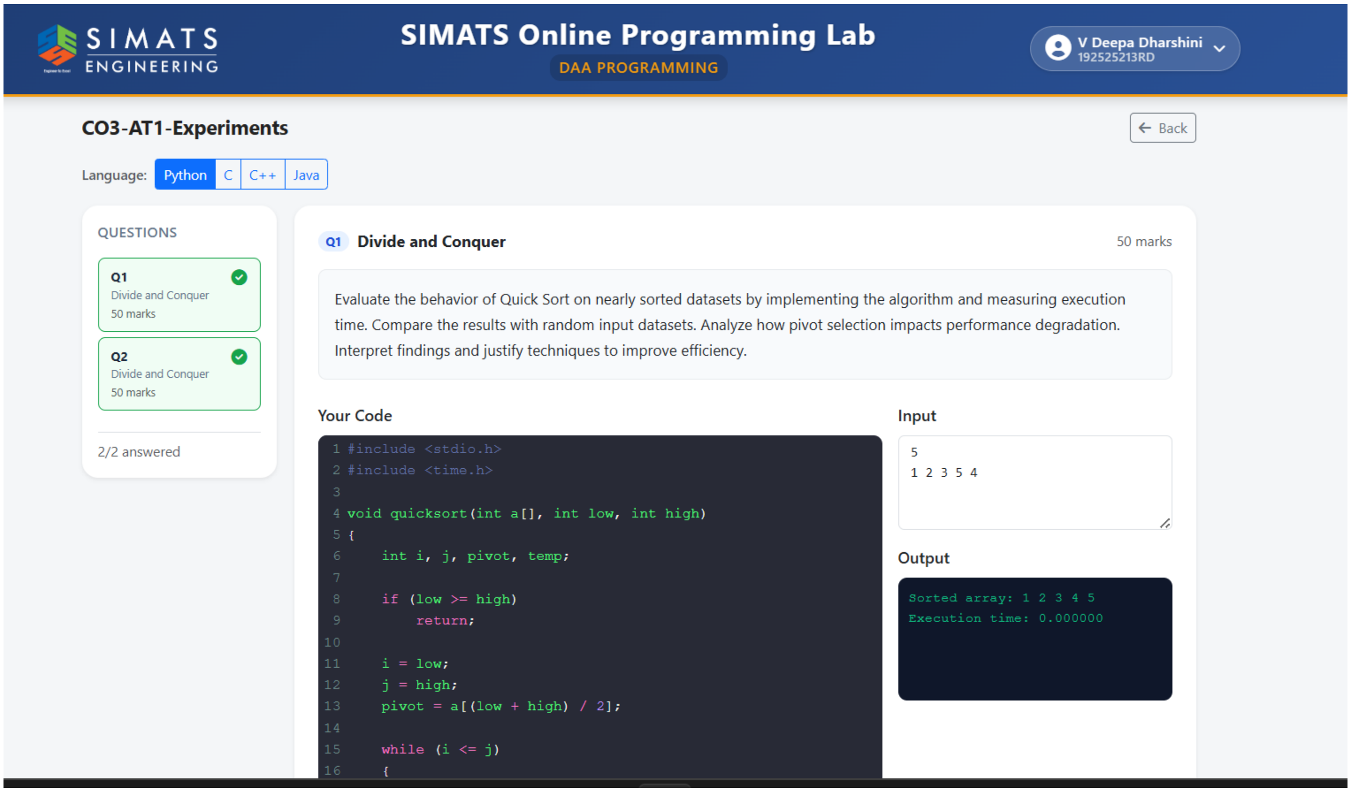
### Instructions to Students

- Write algorithm and execute code both experiments.
- Each questions 50 Marks. Total: 100 Marks.
- Follow a well-structured, systematic approach to design and implement an optimal solution.
- Provide insightful analysis with accurate validation, supported by clear documentation including diagrams, code, and results.

| Question Type | Description | Questions | Total Marks |
|---|---|---|---|
| Experiments | Design and Code for Divide and Conquer Algorithms: Merge Sort, Quick Sort, Binary Search, and Matrix Multiplication | Q1 – Q2 | 100 |

## SECTION A – Experiments






## Code of Conduct

*I V Deepa Dharshini (192525213) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____________________

## RUBRICS FOR CALCULATION

| Experiments | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Weightage** | **Level 4 – Exemplary** | **Level 3 – Proficient** | **Level 2 – Developing** | **Level 1 – Beginning** |
| Understanding of Concepts | 25 | Demonstrates thorough understanding and effectively integrates multiple concepts/technologies | Good understanding with proper integration of most concepts | Basic understanding; limited or partial integration | Poor understanding; unable to integrate concepts |
| Experimental Design | 30 | Well-planned, systematic implementation with optimal solution | Correct implementation with minor issues | Basic implementation with errors or inefficiencies | Incorrect or incomplete implementation |
| Use of Tools | 20 | Effective and advanced use of tools/technologies with proper configuration | Appropriate use of tools with correct execution | Limited or inconsistent use of tools | Incorrect or minimal use of tools |
| Analysis of Results | 15 | Insightful analysis with accurate interpretation and validation of results | Good analysis with reasonable interpretation | Basic analysis with limited insights | Poor or incorrect analysis of results |
| Documentation | 10 | Clear, well-structured documentation with | Organized documentation | Basic documentation | Poor or incomplete |

| | | | | | |
|---|---|---|---|---|---|
| | | diagrams, code, and results | with necessary details | with missing details | documentatio n |

**Marks Evaluation:**

| **Experiments** | Understanding of Concepts **(25)** | Experimental Design **(30)** | Use of Tools **(20)** | Analysis of Results **(15)** | Documentat ion **(10)** | **Total Marks (Each - 50)** |
|---|---|---|---|---|---|---|
| **Ex. 1** | | | | | | |
| **Ex. 2** | | | | | | |
| | | | | | **Overall Total (100)** | **100 %** |

| **Faculty In-charge** | **Course Coordinator** | **Head of Department** |
|---|---|---|
| Name: ___________________<br>Signature: ________________ | Signature: ________________ | Signature: ________________ |
| Date: ____________________ | Date: ____________________ | Date: ____________________ |